^{246}Cm also has a n,γ cross-section of 1.3 b. Because one does not want to loose too much ^{245}Cm, the irradiation is timed to give a maximum yield of ^{246}Cm. If the neutron flux is 2×10^{18} n m^{-2} s^{-1}, (a) when does the ^{246}Cm concentration reach its maximum? (b) What is the ratio between the amount of ^{246}Cm produced and amount of ^{245}Cm consumed at that time?

17.5. A sample containing 50 g ethyl iodide was irradiated with neutrons from a RaBe source of 1×10^9 n m^{-2} s^{-1} for 2 h. If there is a 20% retention and separation occurs 5 min after end of irradiation, what will be the activity of the AgI sample 10 min after separation? The detector has an 8% efficiency for the emitted γ-radiation. Assume the reaction cross-section for ^{127}I(n,γ)^{128}I → (β$^-$, 25.0 min) to be 6.2 b.

17.6. ^{59}Fe of high specific activity can be produced through thermal neutron irradiation of a solution of 1 ml 0.1 M potassium hexacyanoferrate. The recoiling (free) iron atoms, which are produced with a 40% yield, are quantitatively extracted into an organic solvent. How long must the irradiation be in a reactor at 3×10^{19} n m^{-2} s^{-1} to obtain 37 MBq ^{59}Fe? The reaction cross-section for ^{58}Fe(n,γ)^{59}Fe → (β,γ 45.1 d) is 1.15 b and the isotopic abundance of ^{58}Fe is 0.3% in natural iron.

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CHAPTER 18

Uses of Radioactive Tracers

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In this chapter some of the ways in which radiochemistry has aided research in various areas of chemistry and related sciences are reviewed.

The first experiments with radioactive tracers were conducted in 1913 by de Hevesy and Paneth who determined the solubility of lead salts by using one of the naturally occurring radioactive isotopes of lead. Later, after discovery of induced radioactivity,

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